

**GHULAM ISHAQ KHAN INSTITUTE OF ENGINEERING
SCIENCES AND TECHNOLOGY**

ANNUAL ACADEMIC RECORD

(FILLED BY FACULTY MEMBERS)

Year: 2024

NAME HAMZA ABBAS

FACULTY: FME

1. TEACHING:

Semesters - Fall and Spring

[illegible]

*Contact hours 7

Senior Project

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0	Function: T: Teaching
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Summer Training:	
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C: Coordinating

SHORT COURSES AND OTHER TEACHING ACTIVITIES

[illegible]

PART 1: STUDENT EVALUATION:

(Source: Student evaluation in courses, graduating student evaluation, evaluation by participants of short courses, and other sources). Please state your comments, if any, on student evaluation:

Part 2 COURSE MATERIAL COURSE FILE:

(Activities: Organization of course file, development of course material and new courses, adherence to course syllabus, development and use of Innovative methods in teaching, Instructional laboratories, and other activities related to course material development). Please list your activities with regard to course material and comment on them:

- Used “flipped classroom” method in ME 439. The flip flop method promotes active engagement, flexibility, and student-centered learning.

PART 3: OTHER TEACHING ACTIVITIES:

(Advising, supervising senior projects, summer training and students internships, initiation and participation in curriculum development, course co-ordination, and participation in student activities, etc.) Please list and comment on your other teaching activities.

- **Social Media Coordinator FME** – Managing FME Facebook and LinkedIn Page.
- **Batch Advisor (Batch 34) (Non-Lead).**
- **Lab Incharge** – Fluid Mechanics Lab, FME
- Arranged Industrial visit to **DG Khan Cement Company Pvt Ltd.**
- Helped in arranging a student visit to **OGDCL Head Office, Islamabad.**
- Working on feasibility study of Organic Rankine Cycle for **Pakistan Tobacco Company (PTC).**

II RESEARCH AND SCHOLARSHIP

Report Activities for the past academic year only. However, please note that your research profile for the past Two years will be evaluated:

A. RESEARCH ACTIVITIES:

1. Funded Research Project:

Status

Role

Time:Spent

C: Completed
IP: In Progress
T: Terminated

PI: Principal Investigator
CI : Co-Investigator

Percentage of Time as
Approved

Project Title	Funding Agency (Amount in Rs.)	Start Date	End Date	Status	Role	Time Spent

2. Independent Research

Research Title	Start Date	End Date	Hrs/ Wk.	Expected Output and Date Journal/Conference publications, engineering designs, software, technical report etc.
Flow Control in 3D Paper Based Microfluidic Channel	Dec, 2024	Dec, 2025	2.5	1 JP

3. Thesis and Dissertation Supervised

Degree

Role

Msc or Ph.D.

S: Supervisor

CM: Thesis/Dissertation Committee Member

MS and PhD Research

Student Name	Dept.	Deg.	Thesis Title	Start Date	End Date	Role

B: PUBLICATIONS FOR THE LAST TWO YEAR

Please list out by separating into the following:

Class:

JP: Journal Publication TRP: Technical Reports & Proposals
 CP: Conference Publication PPW: Published Professional Work e.g. Engg. Design
 OP: Other Publication PTJ: Publication in Trade journal

Status:

A: Accepted/Published S: Submitted UP: Under Preparation

CLASS	PUBLICATIONS (FOR THE PAST TWO YEARS ONLY)	STATUS
JP	Hamza Abbas , Ali Turab Jafry*, Mubashar Ali, Taqi Ahmad Cheema, “Impact on flow dynamics of paper thickness, contact area, and fluid volume in paper microfluidics (Submitted in: Small Wiley)”. (2024-2025)	S
JP	Mubashar Ali, Ali Turab Jafry*, Hamza Abbas , Minki Lee, Muhammad Salman Abbasi, Syed Farhad Shah, Huma Ajab, Muhammad Kamran, Ali Haider Jafry. “Multiphysics of highly resistive flow in paper microfluidics employing 2D geometry, tortuous path, and adhesive-based enclosed boundary condition”. (2024-2025)	S
CP	Abbas, H. , Ali, M., Naeem, N., Ullah, H., Ali, M., & Jafry, A. T. (2023). Effect of Storage Reservoir on Fluid Velocity in Lateral Flow Paper Device. <i>Engineering Proceedings</i> , 45(1), 5	A
CP	Abbas, H. , Ali, M., Ullah, H., & Jafry, A. T. (2023). Actuation and Flow Control in Paper-Based Microfluidics by Varying Thickness of Storage Reservoir. In <i>MATEC Web of Conferences</i> (Vol. 381, p. 01010). EDP Sciences.	A
CP	Ali, M., Jafry, A. T., Abbas, H. , & Ullah, H. (2023). Flow control in paper-based microfluidic channel by using hydrophobic tape barrier on the paper surface. In <i>MATEC Web of Conferences</i> (Vol. 381, p. 01009). EDP Sciences.	A
CP	Ali, M., Hussain, G., Abbas, H. , Ullah, H., & Jafry, A. T. (2022). Flow control in paper-based microfluidics using Variable porosity channel. <i>Engineering Proceedings</i> , 23(1), 33	A
CP	Ullah, H., Ali, M., Naeem, N., Iqbal, F., Abbas, H. , Jafry, A. T., & Ajab, H. (2024). Droplet Actuation Enhancement through Voltage Control and Hydrophobic Coating Selection. In <i>MATEC Web of Conferences</i> (Vol. 398, p. 01020). EDP Sciences	A
CP	Ali, M., Ullah, H., Ali, M., Naeem, N., Abbas, H. , & Jafry, A. T. (2023). Dynamic Contact Angle Variation with Applied Voltage and Droplet Volume in Digital Microfluidics. <i>Engineering Proceedings</i> , 45(1), 31.	A

CP	Ullah, H., Shoaib, F., Zahid, S. D., Mahmood, M. I., Ali, M., Ali, M., Abbas, H. , & Jafry, A. T. (2023). Design and Fabrication of Digital Microfluidics device for Lab-on-a-Chip Applications. <i>Engineering Proceedings</i> , 45(1), 7.	A
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III COMMITTEES AND OTHER ACTIVITIES (Contribution to meet the Objectives of the Institute)

A: COMMITTEE WORK

FORMED BY: POSITION TYPE OF COMMITTEE
 F: Faculty M: Member S: Standing
 I: Institute C: Chairman a: Ad-Hoc
 G: Other (state)

Name of Committee	Semester	Academic Year	Formed By	Position	Type	App. No of Attended meetings
CTD Committee	Fall 2024	2024	Dr. Ghulam Abbas FCSE	M	I	2
Faculty Study Board	Fall 2024	2024	Dr. Taqi Ahmad Cheema (Dean FME)	M	F	1
Industrial Advisory Board	Fall 2024	2024	Dr. Taqi Ahmad Cheema (Dean FME)	M	F	1

Please describe your major activities and contribution to each committee (Use additional sheets, if necessary). Supervision and mentoring of students, co-curricular and extra-curricular activities and other programs, Student Advisor and Student Society Advisor, any other contribution to meet the Objectives of the Institute.

- **CTD Committee** – CTD Paper Checking.
- **Faculty Study Board** – Suggested few courses title changes & noting Minutes of the Meeting.
- **Industrial Advisory Board** – Noting Minutes of the Meeting.

IV. KNOWLEDGE EXCHANGE

A TECHNICAL SEMINARS AND PRESENTATIONS:

Title	Date	Place
Erasmus+ Training session: Funding your vision [Project Types and Proposal Development]	6 th Dec, 2024	GIK Institute / Online
Faculty Training Session – Fall 2024	2 nd Sept, 2024 – 13 th Sept, 2024	FEE Conference Room, GIK Institute

B PROFESSIONAL CONSULTING:

Organization	Nature of Work	Hrs./Wk	Start Date	End Date

C OTHER PROFESSIONAL ACTIVITIES:

(List patents received, works reviewed, funding received, international and national linkages, new initiatives – lab development, curriculum review new program)

V: CPD AND OTHER ACTIVITIES

Please list any other activity you have undertaken during the Year

Date	Activity

The information given in this form is correct, accurate and complete.

Name Hamza Abbas

Signature/and Date:



31/01/25

Faculty: FME